

ESP32-S3-DualEye-LCD-1.28

From Waveshare Wiki

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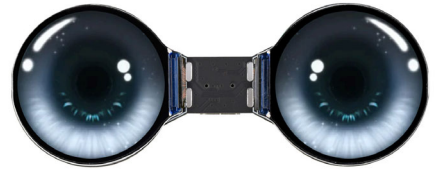
Introduction

[[Expand](#)]

Usage Guide

ESP32-S3-DualEye-LCD-1.28 currently provides two development tools and frameworks, **Arduino IDE** and **ESP-IDF**, providing flexible development options, you can choose the right development tool according to your

ESP32-S3-DualEye-LCD-1.28



(<https://www.waveshare.com/esp32-s3-dualeye-touch-lcd-1.28.htm?sku=32267>)

240×240, UART/SPI

project needs and personal habits.

Development Tool



(/wiki/File:180px-Arduino-IDE-logo.jpg)

Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started. After a simple learning, you can start to develop quickly. At the same time, Arduino has a large global user community, providing an abundance of open source code, project examples and tutorials, as well as rich library resources, encapsulating complex functions, allowing developers to quickly implement various functions.

ESP-IDF



(/wiki/File:180px-ESP-IDF-logo.jpg)

ESP-IDF, or full name Espressif IDE, is a professional development framework introduced by Espressif Technology for the ESP series chips. It is developed using the C language, including a compiler, debugger, and flashing tool, etc., and can be developed via the command lines or through an integrated development environment (such as Visual Studio Code with the Espressif IDF plugin). The plugin offers features such as code navigation, project management, and debugging, etc.

Each of these two development approaches has its own advantages, and developers can choose according to their needs and skill levels. Arduino are suitable for beginners and non-professionals because they are easy to learn and quick to get started. ESP-IDF is a better choice for developers with a professional background or high performance requirements, as it provides more advanced development tools and greater control capabilities for the development of complex projects.

Components Preparation

- ESP32-S3-DualEye-LCD-1.28 x1
- TF card with MP3 files x1

- USB cable (Type-A male to Type-C male) x1

Precautions

- Please note that the computer username must be in English, the username in Chinese will lead to compilation errors
- If you want to continue using the onboard Bluetooth device when enabling all onboard devices with the example provided, it is recommended to use a different audio library file with a small memory footprint (otherwise it will result in a large SRAM footprint)
- **The development board uses USB to download the demo. If the port cannot be recognized, please enter Boot mode (press and hold the Boot button, then connect to the computer, and then release the boot button). After downloading the demo, disconnect the device and reconnect it to the computer to run the demo.**
- The ESP32 3.0.2 on Arduino is based on ESP-IDF v5.1, which is quite different from the previous version based on ESP-IDF V4.X. After the following operations, the original demo may need to be adjusted for normal operation.

Before operating, it is recommended to browse the table of contents to quickly understand the document structure. For smooth operation, please read the FAQ carefully to understand possible problems in advance. All resources in the document are provided with hyperlinks for easy download.

Working with Arduino

This chapter introduces setting up the Arduino environment, including the Arduino IDE, management of ESP32 boards, installation of related libraries, program compilation and downloading, as well as testing demos. It aims to help users master the development board and facilitate secondary development.



(/wiki/File:Arduino-flow-04.png)

Environment Setup

Download and Install Arduino IDE

- Click to visit the Arduino official website (<https://www.arduino.cc/en/software>), select the corresponding system and system bit to download



Arduino IDE 2.3.3

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

SOURCE CODE

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

DOWNLOAD OPTIONS

Windows Win 10 and newer, 64 bits

Windows MSI installer

Windows ZIP file

Linux AppImage 64 bits (X86-64)

Linux ZIP file 64 bits (X86-64)

macOS Intel, 10.15: "Catalina" or newer, 64 bits

macOS Apple Silicon, 11: "Big Sur" or newer, 64 bits

[Release Notes](#)

(/wiki/File:ESP32-S3-AMOLED-1.91-Ar-software-01.png)

- Run the installer and install all by default

The environment setup is carried out on the Windows 10 system, Linux and Mac users can access Arduino-esp32 environment setup (<https://docs.espressif.com/projects/arduino-esp32/en/latest/installing.html>) for reference. (To use on Mac, you need to install the MAC driver (https://files.waveshare.com/wiki/common/CH34XSER_MAC.7z).)

Install ESP32 Development Board

- Before using ESP32-related motherboards with the Arduino IDE, you must first install the software package for the **esp32 by Espressif Systems** development board
- According to **board installation requirement**, it is generally recommended to use **Install Online**. If online installation fails, use **Install Offline**.
- For the installation tutorial, please refer to Arduino board manager tutorial (https://www.waveshare.com/wiki/Arduino_Board_Managers_Tutorial)

- ESP32-S3-DualEye-LCD-1.28 required development board installation description

Board name	Board installation requirement	Version number requirement
esp32 by Espressif Systems	"Install Offline" / "Install Online"	3.2.0 and above

Install Library

- When installing Arduino libraries, there are usually two ways to choose from: **Install online** and **Install offline**. **If the library installation requires offline installation, you must use the provided library file**
For most libraries, users can easily search and install them through the online library manager of the Arduino software. However, some open-source libraries or custom libraries are not synchronized to the Arduino Library Manager, so they cannot be acquired through online searches. In this case, users can only manually install these libraries offline.
- For library installation tutorial, please refer to Arduino library manager tutorial (https://www.waveshare.com/wiki/Arduino_Library_Manager_Tutorial)
- ESP32-S3-DualEye-LCD-1.28 library file is stored in the demo, click here to jump: [ESP32-S3-DualEye-LCD-1.28 Demo](#)
- ESP32-S3-DualEye-LCD-1.28 library file installation instructions

Library Name	Description	Version	Library Installation Requirement
LVGL	Graphical library	v8.3.10	"Install Offline"
ESP32-audioI2S-master	Audio decoding library	v2.0.0	"Install Offline"
es7210	Audio driver library	xxx	"Install Offline"
es8311	Audio driver library	xxx	"Install Offline"

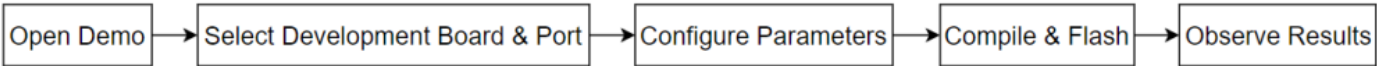
For more learning and use of LVGL, please refer to official documentation (<https://docs.lvgl.io/master/intro/introduction/index.html>)

Run the First Arduino Demo

If you are just getting started with ESP32 and Arduino, and you don't know how to create, compile, flash, and run Arduino ESP32 programs, then please expand and take a look. Hope it can help you!

[\[Expand\]](#)

Demo



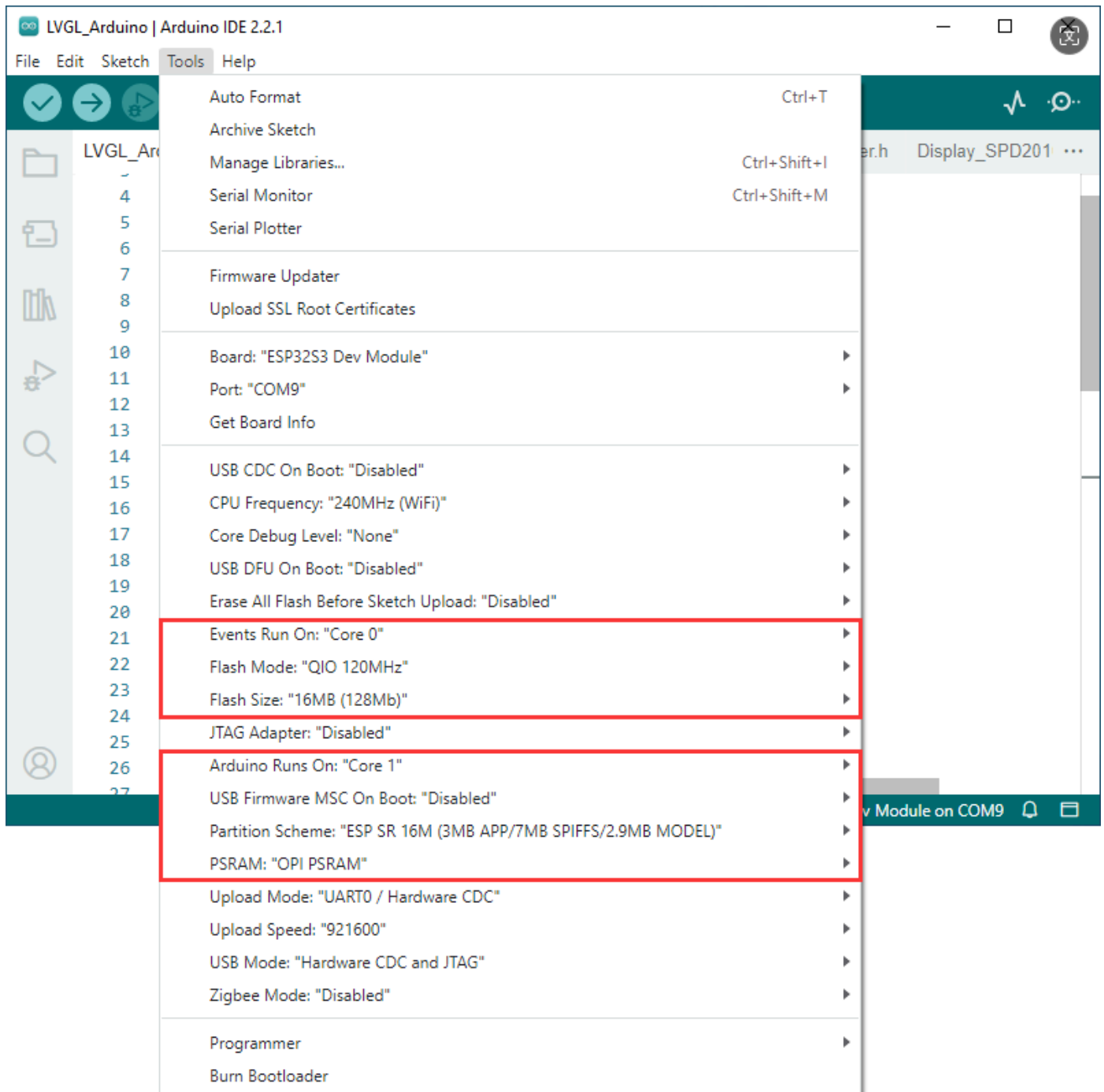
(/wiki/File:Demo-flow-01.png)

- ESP32-S3-DualEye-LCD-1.28 demo

Demo	Basic Description	Dependency Library
LVGL_Arduino	Test onboard device functionality	LVGL, ESP32-audioI2S-master

Arduino Project Parameter Setting

- If the flashed demo **has** a speech recognition model, then select "**ESP SR 16M (3MB APP/7MB SPIFFS/2.9MB MODEL)**" for Partition Scheme
- If the flashed demo **doesn't have** a speech recognition model, then select "**16M Flash (3MB APP/9.9MB FATFS)**" or other for Partition Scheme



(/wiki/File:ESP32-S3-Touch-LCD-1.46_Demo_2.png)

LVGL_Arduino

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Demo description

- This demo demonstrates the functions of each device on the board. Screen 2 displays SD Card, Flash Size, Backlight brightness and other parameters, while screen 1 shows the music playback interface. In addition, this demo includes voice recognition function, which can only be used before playing audio

Hardware connection

- Insert the TF card into the development board
- Connect the development board to the computer

Code analysis

- **setup():** Initialize hardware, configure peripherals, etc.
 - Hardware initialization: `setup()` internally executes multiple initialization functions to configure and initialize hardware components connected to Arduino, such as LCD display, TF card, audio module, microphone, etc. With these initializations, it is ensured that the hardware is working properly
 - Graphics interface initialization: Call `Lvgl1_Init()` to initialize LVGL (Light and Versatile Graphics Library), a widely used graphics library for embedded systems. `Lvgl1_Example1()` is a demo that shows how to render a graphical interface on a display
- **loop():** Perform a certain delay to prevent triggering the watchdog

Result demonstration

- LCD screen display
- LCD screen display parameter description

Parameter	Function	Description
SD Card	Display TF card size	Connect the TF card, if the recognition fails, please format the TF card to FAT32 format (please wait for a while to reset and check again if the recognition fails for the first time)
Flash Size	Display Flash size	Current onboard 16MB Flash
Backlight brightness	Brightness slider	Adjust screen brightness

- Screen 1 is the UI page for playing mp3 audio in the root directory of the TF card
- The UI interface can be controlled by using the BOOT button: click to toggle the selection, double-click to confirm the selection, and long press to switch the screen focus
- This demo enables speech recognition by default, the wake-up word is "hi esp", after waking up, you can speak the command after the backlight is dimmed (if the backlight is not dimmed, it means that it has not been woken up, the recognition requirements are strict, the pronunciation needs to be standard, and the speech speed is slowed down)
- The following are available in several formats of MIC test audio (https://files.waveshare.com/wiki/common/MIC_Test_audio.zip) (Please note that every time the hi esp wake-up fails, please reset the audio to the playback location of the wake-up word and replay it)

- Please do not perform speech recognition when using the speaker to play audio
- The reason the wake-up word plays twice in the test audio is due to the current firmware version's initial wake-up requiring the device to be focused (an analogy), and the current firmware does not permit the disabling of this function

```
// Commands  
Turn on the backlight  
Turn off the backlight  
Backlight is brightest  
Backlight is darkest
```

- Note: The voice recognition feature is only available before playing the audio in the current demo

Switch to Chinese/English recognition model

The initial environment setting is for English recognition, and you can switch to Chinese recognition model or back to English recognition model by following these steps

Switch to Chinese recognition model

[\[Expand\]](#)

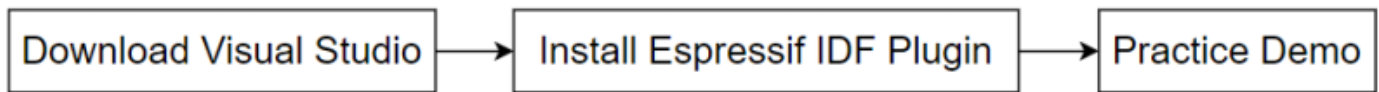
Switch to English recognition model

[\[Expand\]](#)

Working with ESP-IDF

This chapter introduces setting up the ESP-IDF environment setup, including the installation of Visual Studio and the Espressif IDF plugin, program compilation, downloading, and testing of demos, to assist users in mastering the development board and facilitating

secondary development.



(/wiki/File:ESP-IDF-flow-01.png)

Environment Setup

Download and Install Visual Studio

- Open the download page of VScode official website (<https://code.visualstudio.com/download>), choose the corresponding system and system bit to download

Download Visual Studio Code
Free and built on open source. Integrated Git, debugging and extensions.

Windows
Windows 10, 11

User Installer: x64, Arm64
System Installer: x64, Arm64
.zip: x64, Arm64
CLI: x64, Arm64

Linux

.deb
Debian, Ubuntu

.rpm
Red Hat, Fedora, SUSE

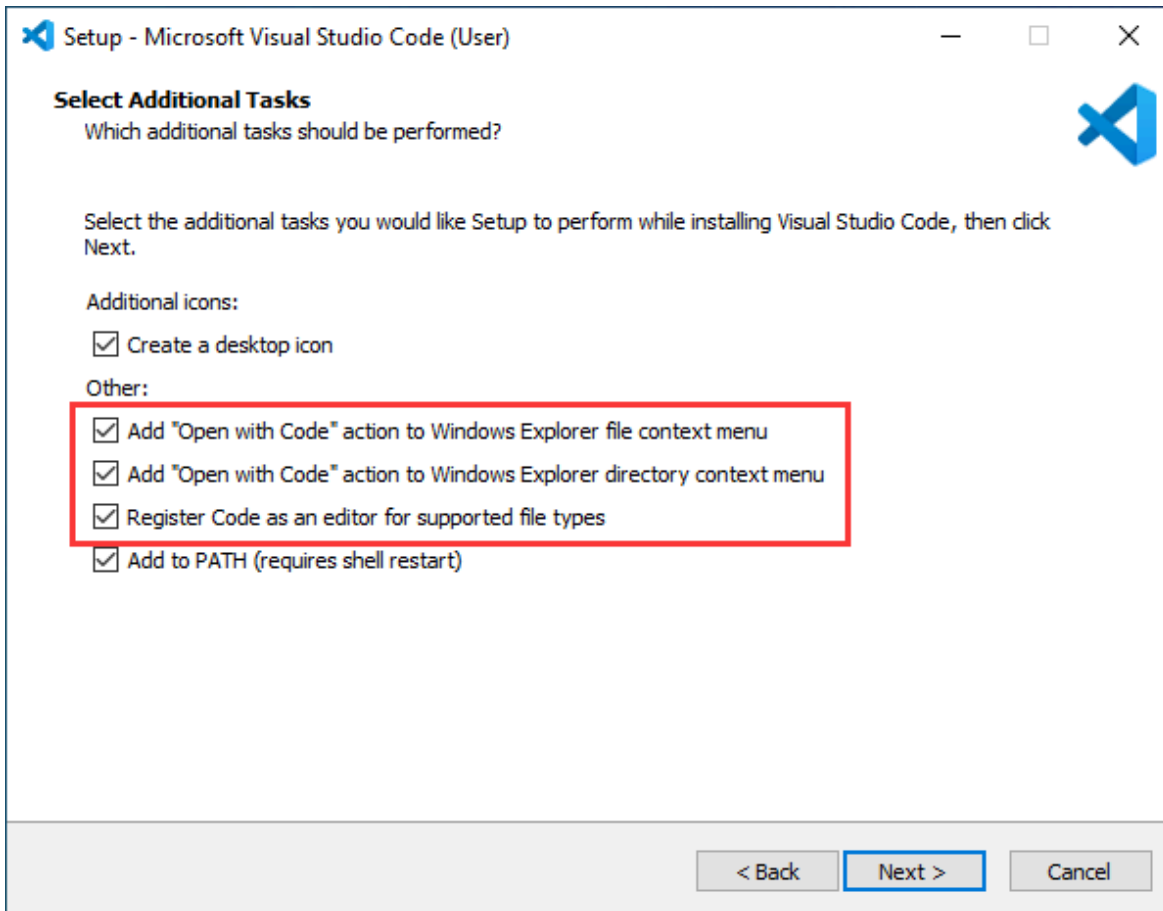
.deb: x64, Arm32, Arm64
.rpm: x64, Arm32, Arm64
.tar.gz: x64, Arm32, Arm64
Snap: Snap Store
CLI: x64, Arm32, Arm64

Mac
macOS 10.15+

.zip: Intel chip, Apple silicon, Universal
CLI: Intel chip, Apple silicon

(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-01.png)

- After running the installation package, the rest can be installed by default, but here for the subsequent experience, it is recommended to check boxes 1, 2, and 3



(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-02.png)

- After the first two items are enabled, you can open VSCode directly by right-clicking files or directories, which can improve the subsequent user experience.
- After the third item is enabled, you can select VSCode directly when you choose how to open it

The environment setup is carried out on the Windows 10 system, Linux and Mac users can access ESP-IDF environment setup (<https://docs.espressif.com/projects/esp-idf/en/v5.1.4/esp32s3/get-started/windows-setup.html>) for reference

Install Espressif IDF Plugin

- It is generally recommended to use **Install Online**. If online installation fails due to network factor, use **Install Offline**
- For more information about how to install the Espressif IDF plugin, see Install Espressif IDF Plugin (https://www.waveshare.com/wiki/Install_Espressif_IDF_Plugin_Tutorial)

- ESP32-S3-DualEye-LCD-1.28 required Espressif IDF version description

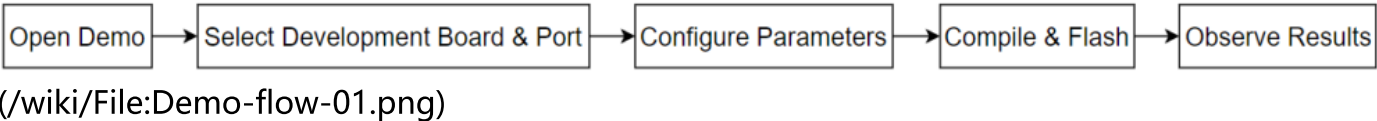
Plugin name	Plugin installation requirement	Version number requirement
Espressif IDF	"Install Offline" / "Install Online"	5.4.1

Run the First ESP-IDF Demo

If you are just getting started with ESP32 and ESP-IDF, and you don't know how to create, compile, flash, and run ESP-IDF ESP32 programs, then please expand and take a look. Hope it can help you!

[Expand]

Demo



- EESP32-S3-DualEye-Touch-LCD-1.28 demo

Demo	Basic Description
01_Text_Number	Test text and number display
02_Shapes_on_Circular	Draw graphics
03_Animated_Eye1	Simulated eye style
04_SD_gif_show	Display GIF images from the TF card
05_Clock	Clock
06_music_player	Test onboard device functionality

[Collapse]

01_Text_Number

Demo description

- This demo demonstrates a text and digital display program on a dual LCD screen, implementing the core function of "dual-screen synchronized display of fixed text + dynamic numbers," cycling through two-digit numbers from 00 to 99, testing LCD screen text and digital display.

Hardware connection

- Connect the development board to the computer

Code analysis

- **Text_Number():** The main interface initialization, obtain the active screen objects of the two displays and create UI
 - Call `create_screen_ui()` to create UI interfaces for two screens, using a vertical layout: text at the top, numbers at the bottom
 - `lv_timer_create(update_numbers, 1000, NULL);` Create a timer with a 1-second interval for number updates (create only once)
- **update_numbers():** Timer callback function, use a timer to call the `update_numbers` function once every 1000 milliseconds (1 second) to update the numbers and refresh the display
 - `(number + 1) % 100;` Incremental numbers (0-99 loop)
 - `lv_label_set_text()` Update the digital label display for two screens

Result demonstration

- LCD screen display



(/wiki/File:900px-01_Text_Number.jpg)

02_Shapes_on_Circular

Demo description

- This demo demonstrates displaying squares, triangles, and circles in random colors sequentially on an LCD display, suitable for learning how to interact with an LCD display using ESP32. Randomly colored squares, triangles, and circles are drawn in the center of the screen one after another, switching every 3 seconds, and testing the stability of graphic drawing.

Hardware connection

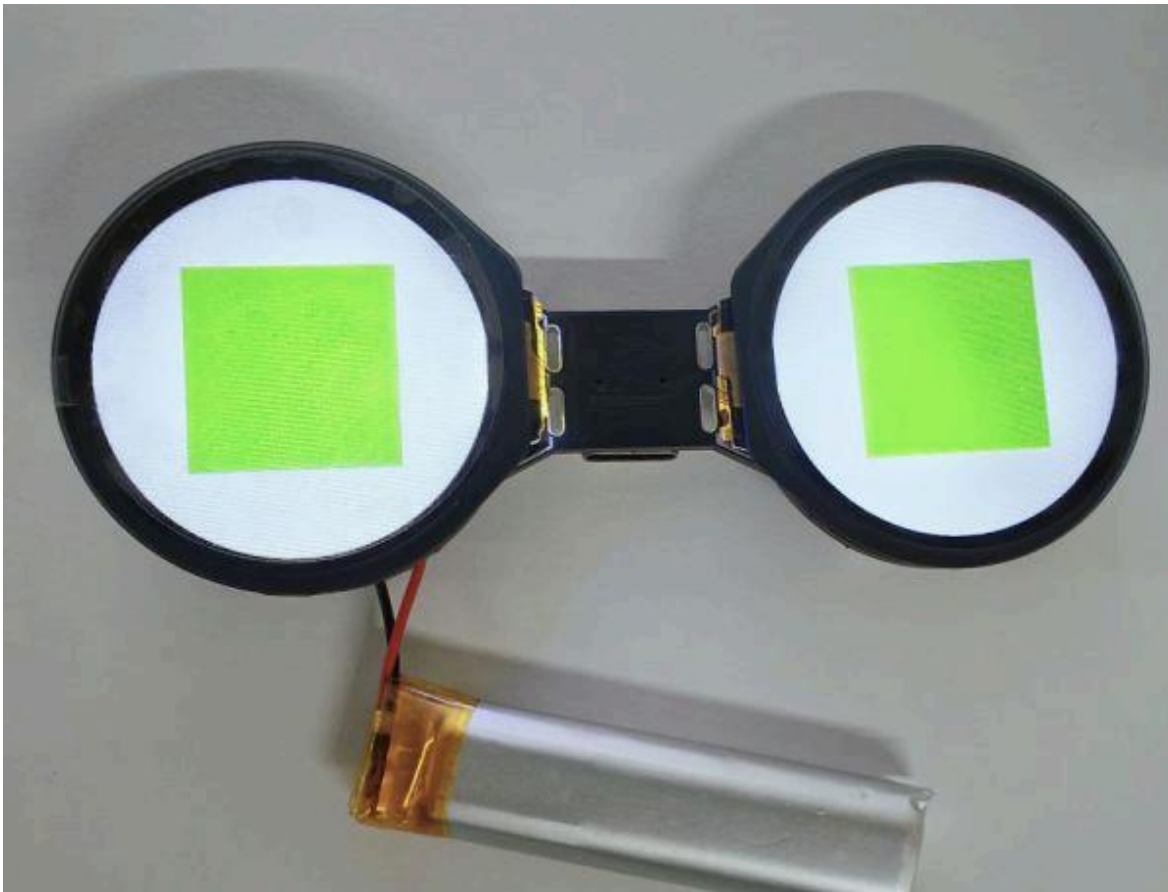
- Connect the development board to the computer

Code analysis

- **Shapes_on_Circular()**: Main initialization function, initialize the random number generator, create the initial shape and start the timer, switching shapes every 3 seconds
- **random_color()**: Generates a random RGB color, providing diverse display effects for shapes
 - Use `rand() % 256` to generate RGB components in the range of 0-255
- **create_base_shape()**: A general shape creation function that can create squares and circles by adjusting the degree of rounding
 - Create an LVGL object and set the specified dimensions, remove the default style, set the background color, opacity, and remove the border
 - Control the degree of rounded corners through the `radius_val` parameter (0=right angle, LV_RADIUS_CIRCLE=circle)

Result demonstration

- LCD screen display



(/wiki/File:900px-02_Shapes_on_Circular_Display.jpg)

03_Animated_Eye1

Demo description

- This example is a dual-screen synchronized dynamic eye animation program based on the LVGL graphics library, achieving real-time rendering and movement logic of "anthropomorphic eyes" — it supports two LCD screens to synchronously display realistic-looking eyes, with the pupils smoothly moving toward random targets to simulate the effect of "natural scanning".

Hardware connection

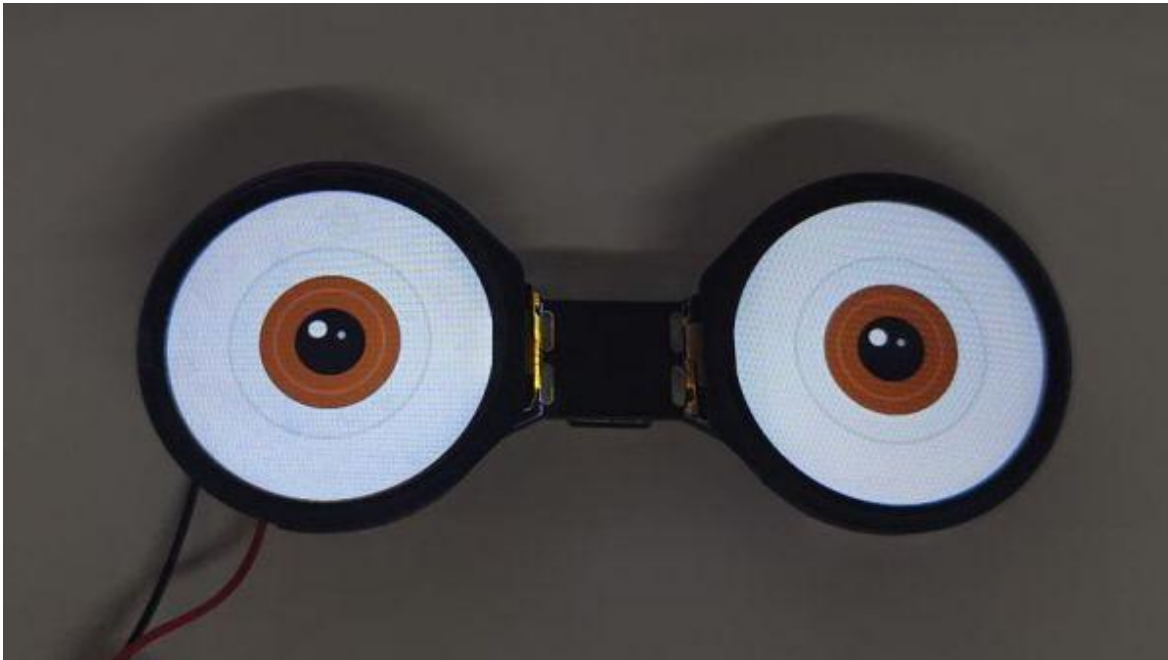
- Connect the development board to the computer

Code analysis

- **Animated_Eye1()**: Program entry point, initializes and starts the eye animation, updates the eye animation effects by creating a timer.
 - Use a fixed random seed `srand(12345)` to ensure that the random behavior is repeatable
 - Create an animation timer with a 50ms interval `lv_timer_create(eye_anim_timer, 50, NULL)`
- **draw_eye()**: Draw the core of the eye, draw the parts of the eye (from outside to inside) in layers to give the eye a sense of realism
 - Calculate sclera (white) fixed center coordinates and radius (apply `SCALE_FACTOR=1.5` scaling)
 - Calculate the actual coordinates of the iris based on the pupil position, limited to the range `MAX_OFFSET_RATIO=0.3`
 - Calculate pupil offset: Additional offset `PUPIL_OFFSET=6` pixels along the iris movement direction
 - Layered drawing: Sclera, iris, pupil, highlight
- **eye_anim_timer()**: Timer callback function, responsible for "updating the target position→ calculating the new position of the pupil→ dual-screen synchronization→ redrawing the eyes" to achieve animation effects

Result demonstration

- LCD screen display



(/wiki/File:900px-03_Animated_Eye1.jpg)

04_SD_gif_show

Demo description

- This example implements the function of synchronously playing animated GIFs on two LCD displays by retrieving .gif images from the root directory of the TF card. The GIF images can be switched using the Boot button, testing the TF card reading and LCD screen image display functions.

Hardware connection

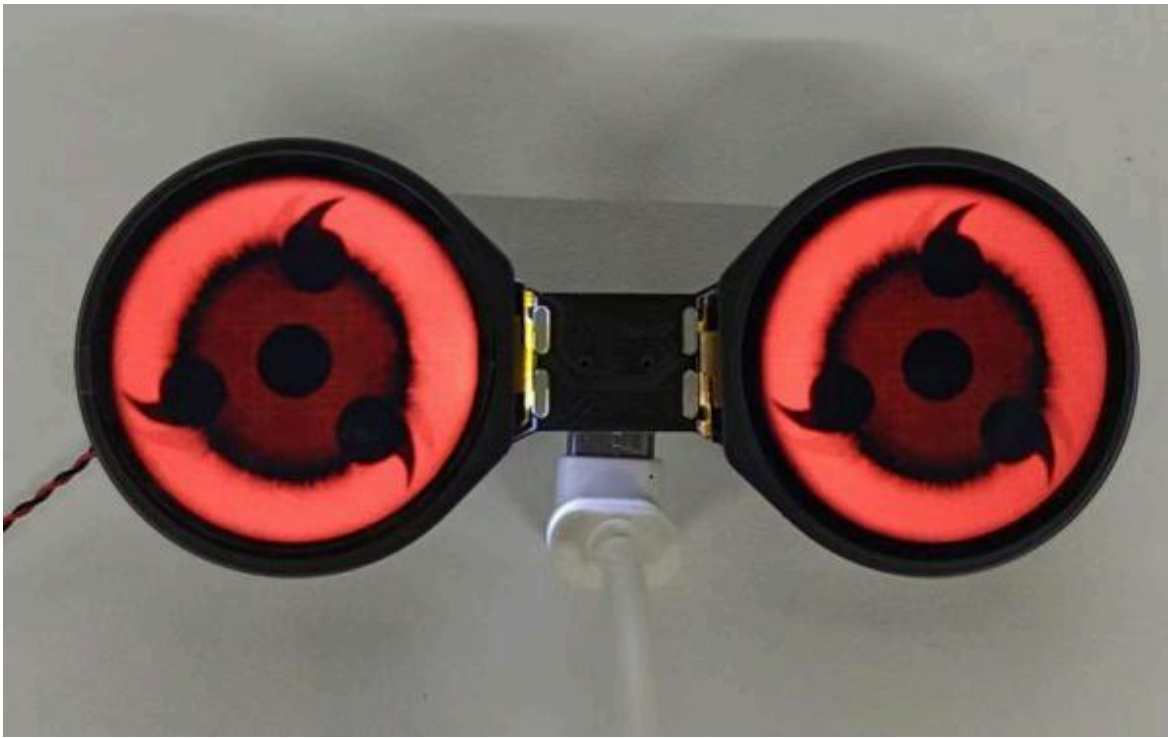
- Copy .gif images in this project directory to the root directory of the TF card (the code and image paths and names can be changed according to your needs. Note: the GIF frame rate needs to be low, otherwise it will be very slow)
- Note: Connect the TF card, if the recognition fails, please format the TF card to FAT32 format (please wait for a while to reset and check again if the recognition fails for the first time)
- Connect the development board to the computer

Code analysis

- **Lvgl_Eyes()**: The main initialization function to create a dual-screen GIF eye display system
 - Initialize FAT file system `lv_fs_fatfs_init()` for TF card access
 - Call `find_gif_files()` to scan GIF files in the root directory of the TF card
 - Create a GIF display object on each screen separately and call `load_current_gif()` to load and play the first GIF file
- **eyes_next_gif()**: Switch to the next gif image.
 - Loop incrementing `current_gif_index` (returning to the first after reaching the end)

Result demonstration

- LCD screen display



(/wiki/File:900px-04_SD_gif_show.jpg)

05_clock

Demo description

- This demo implements a clock function that synchronously displays on two screens, including an analog clock (with pointers and scales) and a digital time display

Hardware connection

- Connect the development board to the computer

Code analysis

- **Clock():** : The main program entrance, initializes the clocks of two LCD displays, sets the initial state, and starts the timer to achieve dynamic updates.
 - Call `create_clock_elements()` separately on two screens to create clock canvases and time labels

- Get the current time and calculate the initial pointer angle
- **get_current_time()**: Time acquisition and processing, obtain the current time from the system
- **draw_clock_ticks()**: Clock scale drawing, drawing 12 scales on the clock canvas, of which 3, 6, 9, 12 are longer and wider main scales, and the rest are minor scales
- **draw_clock()**: Complete clock drawing, integrates the drawing of scales and pointers, generating a complete analog clock image
 - Call `draw_hand()` separately to draw three pointers

Result demonstration

- LCD screen display



(/wiki/File:900px-05_Clock.jpg)

06_music_player

Demo description

- This demo demonstrates the functions of each device on the board. Screen 2 displays SD Card, Flash Size and other parameters, while screen 1 shows the music playback interface. In addition, this demo includes voice recognition function.

Hardware connection

- Insert the TF card into the development board
- Connect the development board to the computer

Code analysis

- **Driver_Init():** Initializes the resources required by the hardware, driver, and system, and creates a task to cycle through the hardware driver
 - Hardware initialization: `Driver_Init()` is responsible for initializing multiple hardware modules, such as I2C bus, sensors, etc.
 - Modular design: The initialization function disperses the initialization of each hardware into different modules, which is easy to maintain and expand. If there's new hardware that needs to be supported, just add the initialization code here
- **Lvgl_Example1():** Display UI on the screen

Result demonstration

- LCD screen display
- LCD screen display parameter description

Parameter	Function	Description
SD Card	Display TF card size	Connect the TF card, if the recognition fails, please format the TF card to FAT32 format (please wait for a while to reset and check again if the recognition fails for the first time)
Flash Size	Display Flash size	Current onboard 16MB Flash
Battery Voltage	Battery voltage	The battery voltage can be detected when the battery is connected
Wireless scan	Display the number of scanned WiFi	Scan completion will display ..OK at the end

- Page 2 is the UI page for playing mp3 audio in the root directory of the TF card
- This demo enables speech recognition by default, the wake-up word is "hi esp", after waking up, you can speak the command after the backlight is dimmed (if the backlight is not dimmed, it means that it has not been woken up, the recognition requirements are strict, the pronunciation needs to be standard, and the speech speed is slowed down)
- The following are available in several formats of MIC test audio (https://files.waveshare.com/wiki/common/MIC_Test_audio.zip) (Please note that every time the hi esp wake-up fails, please reset the audio to the playback location of the wake-up word and replay it)
- Please do not perform speech recognition when using the speaker to play audio

- The reason the wake-up word plays twice in the test audio is due to the current firmware version's initial wake-up requiring the device to be focused (an analogy), and the current firmware does not permit the disabling of this function

```
// Commands  
Turn on the backlight  
Turn off the backlight  
Backlight is brightest  
Backlight is darkest
```

- Note: The voice recognition feature is only available before playing the audio in the current demo



(/wiki/File:1500px-ESP32-S3-DualEye-Touch-LCD-1.28-1.jpg)

Switch to Chinese/English recognition model

The initial environment setting is for English recognition, and you can switch to Chinese recognition model or back to English recognition model by following these

steps

Switch to Chinese/English recognition model

[\[Expand\]](#)

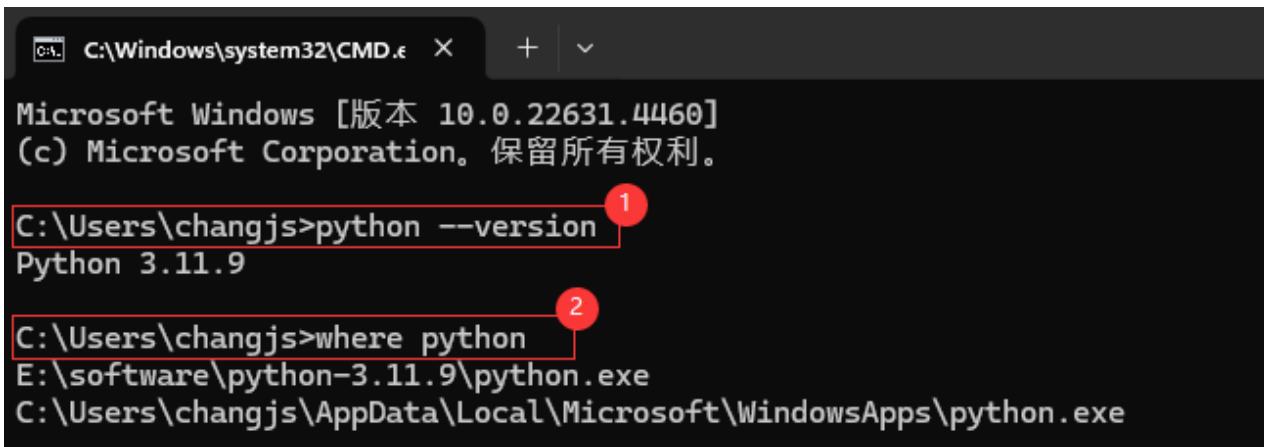
Generate Speech Control Command

Environment Preparation

- Check if your computer has a Python environment
- Open the command line: win+r -> enter cmd

①. Enter `python --version` and the python version number appears, which means that it is installed

②. Type `where python` to see where python is installed



```
C:\Windows\system32\CMD.exe X + v
Microsoft Windows [版本 10.0.22631.4460]
(c) Microsoft Corporation。保留所有权利。
C:\Users\changjs>python --version 1
Python 3.11.9
C:\Users\changjs>where python 2
E:\software\python-3.11.9\python.exe
C:\Users\changjs\AppData\Local\Microsoft\WindowsApps\python.exe
```

(/wiki/File:Python-install-04.png)

The Python environment under the C drive is the Python environment that is automatically installed when VS Code is installed;

The disk E contains the Python environment used in the following steps;

If you do not have installed environment, please refer to Python Environment Installation Tutorial (https://www.waveshare.com/wiki/Install_Python_Environment_Tutorial), please continue if it is installed

- On the command line, enter **`pip install g2p_en`** to install the g2p_en package
- Install the resource package (https://files.waveshare.com/wiki/common/nltk_data.exe) in the corresponding user's **`C:\Users\username\AppData\Roaming`** directory, and double-click to extract the file

- On the command line, enter **pip install pypinyin** to install the pypinyin library

Generate Chinese Pinyin

- Download the Chinese command generation file (https://files.waveshare.com/wiki/common/Generate_Chinese_command.zip)
- Command line switches the path to **Chinese command generation folder** (the path varies from person to person, the following is only for reference)

```
cd /d E:\download\Generate_Chinese_command
```

- Execute the command line **multinet_pinyin.py** to generate Chinese Pinyin

```
python multinet_pinyin.py "Chinese to be converted to Pinyin"
Example: Python multinet_pinyin.py "你好微雪电子"
```

```
E:\download\Generate_Chinese_command>python multinet_pinyin.py "你好微雪电子"
enum {
    SR_CMD_NI_HAO_WEI_XUE_DIAN_ZI,
};

static const sr_cmd_t sr_commands[] = {
    { 0, "ni hao wei xue dian zi", "ni hao wei xue dian zi" },
};

E:\download\Generate_Chinese_command>
```

(/wiki/File:ESP32-S3-Touch-LCD-1.85-demo-02.png)

Generate English Phonemes

- Download the English command generation file (https://files.waveshare.com/wiki/common/Generate_English_command.zip)
- Command line switch path to **English command generation folder** (the path varies from person to person, the following is only for reference)

```
cd /d E:\download\Generate_English_command
```

- Execute the command line **gen_sr_commands.py** to generate English phonemes

```
python gen_sr_commands.py "English to be converted to phonemes"
Example: python gen_sr_commands.py "hello waveshare"
```

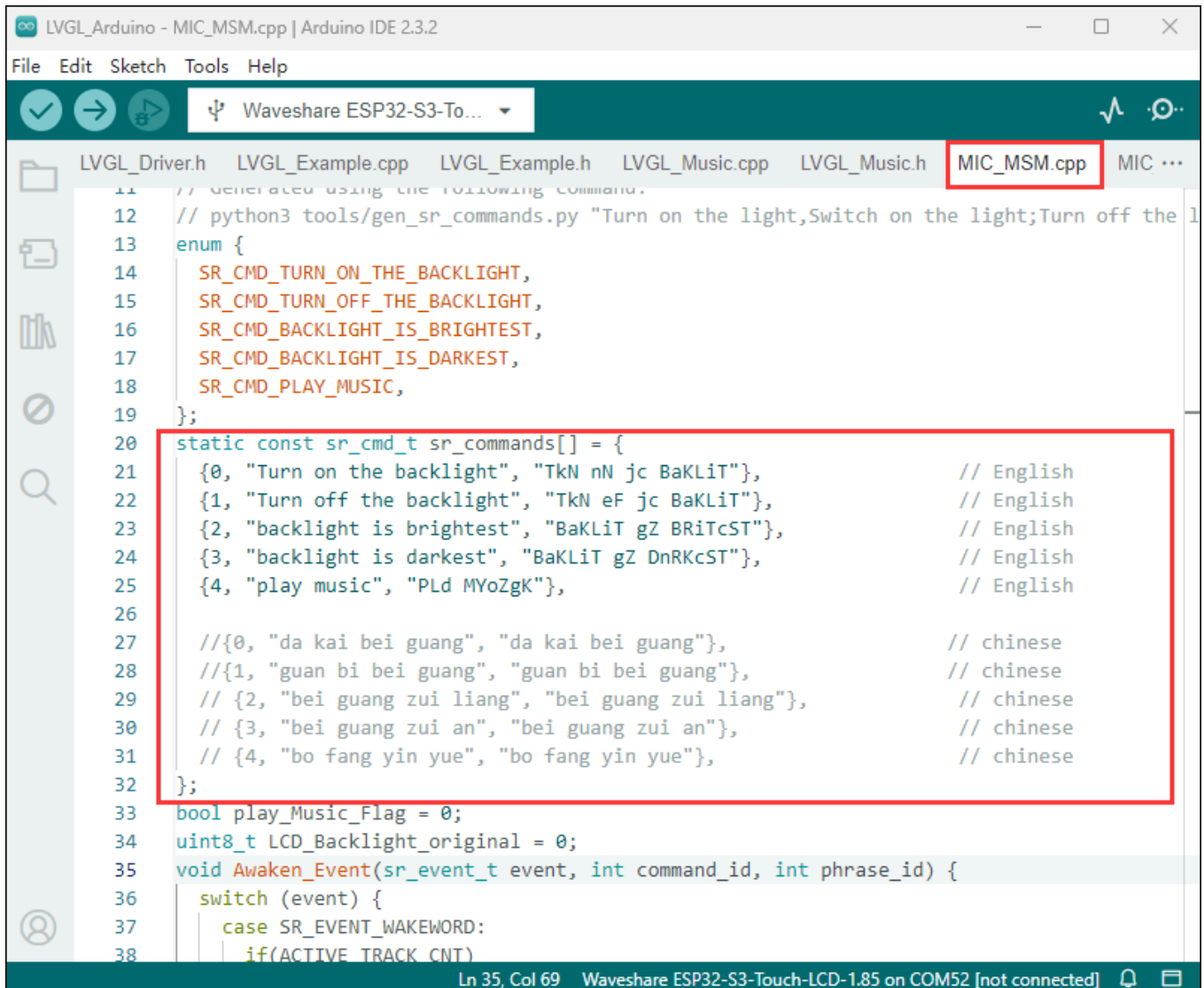


```
E:\download\Generate_English_command>python gen_sr_commands.py "hello waveshare"
enum {
    SR_CMD_HELLO_WAVESHARE,
};
static const sr_cmd_t sr_commands[] = {
    { 0, "hello waveshare", "hcLb WdVsnR"},
};
E:\download\Generate_English_command>
```

(/wiki/File:ESP32-S3-Touch-LCD-1.85-demo-03.png)

Command modification

In Arduino



```
LVGL_Arduino - MIC_MSM.cpp | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Waveshare ESP32-S3-To...
LVGL_Driver.h LVGL_Example.cpp LVGL_Example.h LVGL_Music.cpp LVGL_Music.h MIC_MSM.cpp MIC ...
11 // generated using the following command.
12 // python3 tools/gen_sr_commands.py "Turn on the light,Switch on the light;Turn off the light"
13 enum {
14     SR_CMD_TURN_ON_THE_BACKLIGHT,
15     SR_CMD_TURN_OFF_THE_BACKLIGHT,
16     SR_CMD_BACKLIGHT_IS_BRIGHTEST,
17     SR_CMD_BACKLIGHT_IS_DARKEST,
18     SR_CMD_PLAY_MUSIC,
19 };
20 static const sr_cmd_t sr_commands[] = {
21     {0, "Turn on the backlight", "Tkn nN jc BaKLiT"}, // English
22     {1, "Turn off the backlight", "Tkn eF jc BaKLiT"}, // English
23     {2, "backlight is brightest", "BaKLiT gZ BRiTcST"}, // English
24     {3, "backlight is darkest", "BaKLiT gZ DnRKcST"}, // English
25     {4, "play music", "PLd MYoZgK"}, // English
26
27     //{0, "da kai bei guang", "da kai bei guang"}, // chinese
28     //{1, "guan bi bei guang", "guan bi bei guang"}, // chinese
29     //{2, "bei guang zui liang", "bei guang zui liang"}, // chinese
30     //{3, "bei guang zui an", "bei guang zui an"}, // chinese
31     //{4, "bo fang yin yue", "bo fang yin yue"}, // chinese
32 };
33 bool play_Music_Flag = 0;
34 uint8_t LCD_Backlight_original = 0;
35 void Awaken_Event(sr_event_t event, int command_id, int phrase_id) {
36     switch (event) {
37         case SR_EVENT_WAKEWORD:
38             if(ACTIVE_TRACK CNT)
```

(/wiki/File:ESP32-S3-Touch-LCD-1.85-demo-06.png)

In ESP-IDF

- Chinese commands

The screenshot shows the Visual Studio Code editor with the file explorer on the left. The file explorer shows the project structure for 'ESP32-S3-TOUCH-LCD-1.85-Test'. The file 'sdkconfig.defaults' is selected and highlighted with a red box. The main editor window displays the contents of 'sdkconfig.defaults', which includes Chinese speech commands for the ESP32-S3. The commands are defined as CONFIG_CN_SPEECH_COMMAND_ID0 through CONFIG_CN_SPEECH_COMMAND_ID11. The status bar at the bottom shows 'ESP-IDF v5.3.1', 'UART', 'COM52', 'esp32', and 'Build'.

```

97 #
98 # Add Chinese speech commands
99 #
100 CONFIG_CN_SPEECH_COMMAND_ID0="bei guang zui liang"
101 CONFIG_CN_SPEECH_COMMAND_ID1="bei guang zui an"
102 CONFIG_CN_SPEECH_COMMAND_ID2="guan bi bei guang"
103 CONFIG_CN_SPEECH_COMMAND_ID3="da kai bei guang"
104 CONFIG_CN_SPEECH_COMMAND_ID4="bo fang yin yue"
105 CONFIG_CN_SPEECH_COMMAND_ID5=""
106 CONFIG_CN_SPEECH_COMMAND_ID6=""
107 CONFIG_CN_SPEECH_COMMAND_ID7=""
108 CONFIG_CN_SPEECH_COMMAND_ID8=""
109 CONFIG_CN_SPEECH_COMMAND_ID9=""
110 CONFIG_CN_SPEECH_COMMAND_ID10=""
111 CONFIG_CN_SPEECH_COMMAND_ID11=""

```

(/wiki/File:ESP32-S3-Touch-LCD-1.85-demo-10.png)

English commands

The screenshot shows the Visual Studio Code editor with the file explorer on the left. The file explorer shows the project structure for 'ESP32-S3-TOUCH-LCD-1.85-Test'. The file 'sdkconfig.defaults' is selected and highlighted with a red box. The main editor window displays the contents of 'sdkconfig.defaults', which includes English speech commands for the ESP32-S3. The commands are defined as CONFIG_EN_SPEECH_COMMAND_ID0 through CONFIG_EN_SPEECH_COMMAND_ID12. The status bar at the bottom shows 'ESP-IDF v5.3.1', 'UART', 'COM52', 'esp32', and 'Build'.

```

306 #
307 # Add English speech commands
308 #
309 CONFIG_EN_SPEECH_COMMAND_ID0="BaKLiT gZ BRiTcST" # Backlight is brightest
310 CONFIG_EN_SPEECH_COMMAND_ID1="BaKLiT gZ DnRKcST" # Backlight is darkest
311 CONFIG_EN_SPEECH_COMMAND_ID2="TkN eF jc BaKLiT" # Turn off the backlight
312 CONFIG_EN_SPEECH_COMMAND_ID3="TkN nN jc BaKLiT" # Turn on the backlight
313 CONFIG_EN_SPEECH_COMMAND_ID4="PLd MYoZgK" # Play music
314 CONFIG_EN_SPEECH_COMMAND_ID5=""
315 CONFIG_EN_SPEECH_COMMAND_ID6=""
316 CONFIG_EN_SPEECH_COMMAND_ID7=""
317 CONFIG_EN_SPEECH_COMMAND_ID8=""
318 CONFIG_EN_SPEECH_COMMAND_ID9=""
319 CONFIG_EN_SPEECH_COMMAND_ID10=""
320 CONFIG_EN_SPEECH_COMMAND_ID11=""
321 CONFIG_EN_SPEECH_COMMAND_ID12=""

```

(/wiki/File:ESP32-S3-Touch-LCD-1.85-demo-09.png)

Flash Firmware Flashing and Erasing

- The current demo provides test firmware, which can be used to test whether the onboard device functions properly by directly flashing the test firmware
- bin file path:

..\ESP32-S3-DualEye-LCD-1.28-Demo\Firmware

Flash firmware flashing and erasing (https://www.waveshare.com/wiki/Flash_Firmware_Flashing_and_Erasing) for reference

Resources

Schematic Diagram

- ESP32-S3-LCD-1.28 Schematic diagram (<https://files.waveshare.com/wiki/ESP32-S3-DualEye-LCD-1.28/ESP32-S3-DualEye-LCD-1.28-Schematic.pdf>)

Project Diagram

- ESP32-S3-DualEye-LCD-1.28 Structure file (<https://files.waveshare.com/wiki/ESP32-S3-DualEye-LCD-1.28/ESP32-S3-DualEye-LCD-1.28.zip>)

Demo

- ESP32-S3-DualEye-LCD-1.28 Demo (<https://drive.google.com/file/d/18E9IHihcN9UW8ts0qSyKKCYiPRkN6ZZT/view>)

Datasheets

ESP32-S3

- ESP32-S3 Technical Reference Manual (https://files.waveshare.com/wiki/common/Esp32-s3_technical_reference_manual_en.pdf)

- ESP32-S3 Series Datasheet (https://files.waveshare.com/wiki/common/Esp32-s3_datasheet_en.pdf)

Software Tools

Arduino

- Arduino IDE Official download link (<https://www.arduino.cc/en/software/>)

VScode

- VScode Official download link (<https://code.visualstudio.com/download>)

Firmware Flashing Tool

- Flash_download_tool (https://dl.espressif.com/public/flash_download_tool.zip)

Other Resource Links

- ESP32-Arduino official documentation (<https://docs.espressif.com/projects/arduino-esp32/en/latest/index.html>)
- LVGL official documentation (<https://docs.lvgl.io/master/intro/introduction/index.html>)
- MIC test audio (https://files.waveshare.com/wiki/common/MIC_Test_audio.zip)

FAQ

Question: What should I do if the ESP-IDF program keeps resetting after flashing?

Answer:

- Downgrade ESP-IDF version to v5.4.1

Question: After the module downloads the demo and re-downloads it, why sometimes it can't connect to the serial port or the flashing fails?

Answer:

- Click the Reset button for more than 1 second, wait for the PC to re-recognize the device and then download again

- Long press the BOOT button, press RESET at the same time, then release RESET, then release the BOOT button, at this time the module can enter the download mode, which can solve most of the problems that can not be downloaded.
- Press and hold the BOOT button to reconnect the USB cable, and then release the BOOT button after the USB cable is connected, at this time the module can enter the download mode, which can solve most of the problems that can not be downloaded.

Question: Why does the module keep resetting and flicker when viewed the recognition status from the device manager?

Answer:

It may be due to Flash blank and the USB port is not stable, you can long-press the BOOT button, press RESET at the same time, and then release RESET, and then release the BOOT button, at this time the module can enter the download mode to flash the firmware (demo) to solve the situation.

Question: How to deal with the first compilation of the program being extremely slow?

Answer:

- It's normal for the first compilation to be slow, just be patient

Question: What should I do if I can't find the AppData folder?

Answer:

- Some AppData folders are hidden by default and can be set to show.
- English system: Explorer->View->Check "Hidden items"
- Chinese system: File Explorer -> View -> Display -> Check "Hidden Items"

Question: How do I check the COM port I use?

Answer:

■ Windows system:

- ①View through Device Manager: Press the Windows + R keys to open the "Run" dialog box; input devmgmt.msc and press Enter to open the Device Manager; expand the "Ports (COM and LPT)" section, where all COM ports and their current statuses will be listed.
- ②Use the command prompt to view: Open the Command Prompt (CMD), enter the "mode" command, which will display status information for all COM ports.
- ③Check hardware connections: If you have already connected external devices to the COM port, the device usually occupies a port number, which can be determined by checking the connected hardware.

■ Linux system:

- ①Use the dmesg command to view: Open the terminal.
- ①Use the ls command to view: Enter ls /dev/ttyS* or ls /dev/ttyUSB* to list all serial port devices.
- ③Use the setserial command to view: Enter setserial -g /dev/ttyS* to view the configuration information of all serial port devices.

Question: Why does ESP-IDF report this error?

```

I (2341) gpio: GPIO[2]| InputEn: 0| OutputEn: 0| OpenDrain: 0| Pullup: 1| Pulldown: 0| Intr:0
I (2341) gpio: GPIO[1]| InputEn: 0| OutputEn: 0| OpenDrain: 0| Pullup: 1| Pulldown: 0| Intr:0
I (2351) gpio: GPIO[42]| InputEn: 0| OutputEn: 0| OpenDrain: 0| Pullup: 1| Pulldown: 0| Intr:0
E (2381) sdmmc_common: sdmmc_init_ocr: send_op_cond (1) returned 0x107
E (2381) vfs_fat_sdmmc: sdmmc_card_init failed (0x107).
E (2381) SD: Failed to initialize the card (ESP_ERR_TIMEOUT). Make sure SD card lines have pull-up resistors in place.
I (2401) LVGL: Initialize LVGL library
I (2401) LVGL: Use frame buffers as LVGL draw buffers
I (2411) LVGL: Register display driver to LVGL

```

(/wiki/File:ESP32-S3-Touch-LCD-2.1-ESP_IDE-demo-02.png)

Answer:

- This situation is that the TF card is not installed or the TF card cannot be recognized. If this situation occurs, please wait for a period of time to reset the device. If it still cannot be resolved, please format the TF card to FAT32 format. If it still fails, please try a different TF card

Question: Why does the program flashing fail when using a MAC device?**Answer:**

- Install MAC Driver (https://files.waveshare.com/wiki/common/CH34XSER_MAC.7z) and flash again.

Question: What is the screen refresh rate? (Frame rate)**Answer:**

Depending on the demo you run, different demos have different refresh frame rates.

Question: When I run a program (Serial.println()) that uses a serial port, the Serial Monitor in my Arduino IDE does not show any data?**Answer:**

Whether you use the USB to UART tool to connect the UART pins, or enable USB CDC in Tools

Question: Can Wi-Fi and Bluetooth be used simultaneously?**Answer:**

Yes, it can be used simultaneously, but due to the larger memory usage of audio, the audio function cannot be turned on at the same time

Question: Which battery can be used?**Answer:**

MX1.25 interface 3.7V lithium battery

Question: What capacity lithium battery is recommended?**Answer:**

Use the one that corresponds to our interface and output voltage, the capacity depending on customer needs

Question: Why is there no output after successfully flashing the code with no issues?**Answer:**

- Check the schematic diagram for different development boards with Type-C interfaces, and handle the output accordingly:
 - For development boards with direct USB output, printf function is supported for printing output. If you want to support output via the Serial function, you will need to enable the USB CDC On Boot feature or declare HWCDC.
 - For development boards with UART to USB conversion, both printf and Serial functions are supported for printing output, and there is no need to enable USB CDC On Boot.

Question: How to use SquareLine Studio to design interfaces?**Answer:**

- Please refer to SquareLine Studio tutorial (https://www.waveshare.com/wiki/Waveshare_SquareLine_Studio)

Question: How to restore factory mode?**Answer:**

- Flash the factory bin file, refer to Flash Firmware Flashing and Erasing

Question: Can I use our product without a TF card?**Answer:**

OK, just the functions related to TF card cannot be used, and other functions are not affected

Question: Screen issue**Answer:**

The screen and back panel of this type of product adopt an integrated bonding design. To ensure normal use, it is recommended to avoid disassembling or prying open the adhesive layer by yourself. If the screen is damaged due to disassembly, it cannot be repaired

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8
(Monday to Friday)

Submit Now (<https://service.waveshare.com/>)

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(<https://www.waveshare.com/w/index.php?title=ESP32-S3-DualEye-LCD-1.28&oldid=108407>)"*
